Resource Allocation for Multi-Modal Semantic Communication in UAV Collaborative Networks

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***Abstract*—Semantic communication is envisioned as a potential communication paradigm enabled by artificial intelligence and is promising to break the Shannon limit for future 6G networks. This paradigm benefits uninhabited aerial vehicles (UAVs) to conserve communication resources and minimize latency by only transmitting task-relevant semantic information. However, resource allocation in the multiple collaborative UAV scenarios remains unexplored, particularly regarding multi-modal semantic communication. To tackle this challenge, this paper investigates a semantic-aware intelligent resource allocation method for multi-UAV-assisted semantic communication networks in the UAV image-sensing task-oriented scenario. A multi-modal semantic communication framework with multi-UAV relay collaboration is developed. At the semantic level, a novel quality of experience (QoE) and the transmission cost model are introduced, based on which a semantic-aware resource allocation problem is formulated, aiming to maximize QoE while minimizing the transmission cost by jointly optimizing the UAV trajectory, the spectrum bandwidth, the transmit power and the number of the transmitted semantic symbols. To deal with optimization challenges involving hybrid variables and coordination among UAVs, a multi-UAV hybrid decision-controlled deep reinforcement learning (DRL) scheme is proposed. Simulation results demonstrate the effectiveness of the proposed scheme compared with the benchmark schemes in achieving a good balance between the QoE and the transmission cost.**

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***Index Terms*—uninhabited aerial vehicle (UAV), semantic communication, resource allocation, deep reinforcement learning (DRL).**

I. Introduction

THE rapid advancement of aviation and information technology has catalyzed the emergence of a low-altitude economy, which is a novel economic mode that capitalizes on resources within the airspace below 3,000 meters [1]. Recently, such an economy has consistently bolstered industries such as tourism, transportation, agriculture, emergency rescue, and environmental monitoring, injecting fresh momentum into the national economy. Due to high maneuverability, low energy consumption, and real-time monitoring capabilities, uninhabited aerial vehicles (UAVs) play a vital role and open up new opportunities for the low-altitude economy by offering flexible and cost-efficient services across various fields [2], [3], [4]. However, the massive connectivity of smart IoT devices, coupled with the explosive growth of wireless data traffic, has caused the escalating challenge of the constrained onboard energy in UAV communication networks [5], [6], [7]. Additionally, current communication technologies emphasize accurate transmission at the bit level, while neglecting the value and intrinsic meaning of transmitted data [8]. Furthermore, smart IoT devices often generate a large amount of low-value density data, leading to the unnecessary consumption of communication resources in traditional communication-based UAV networks [9], [10], [11]. As a result, instead of pursuing higher network bit rates of the traditional communication method, it is imperative to explore alternative communication paradigms for UAV communications.

Fortunately, semantic communication has emerged as a promising enabler for 6G and future applications [12]. Unlike traditional bit-level communication, which emphasizes the transmission of raw bits, AI-driven semantic communication focuses on transmitting meaningful content. By extracting and transmitting semantic information, it can significantly reduce unnecessary data transmission. Addtionally, its integration of prior and background knowledge further enhances the accuracy and efficiency of data delivery [13], [14], [15], [16]. Consequently, semantic communication is increasingly seen as a key solution to address the limitations of traditional UAV communication networks [17], [18], [19].

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With the rapid advancement of AI, multi-modal semantic communication has attracted significant attention due to its promising applications, such as the visual question answering (VQA) task, which integrates both text and image information [15]. By fusing and integrating various modalities of semantic feature data [20], [21], the flexibility, robustness, and adaptability of semantic communication can be significantly enhanced. For instance, in emergency rescue scenarios, UAVs equipped with cameras and infrared sensors can improve the accuracy of anomaly detection and situational awareness by fusing visual and thermal information [22]. Additionally, multi-modal transformation enables the flexible conversion of high-data-volume modalities (e.g., video or image) into low-data-volume modalities (e.g., textual descriptions) [23], which helps conserve transmission resources, reduce transmission latency and improve decision-making efficiency.

In the face of resource-constrained challenges, implementing reasonable resource allocation strategies in semantic communication of UAV networks is also critically important [24]. Unlike traditional bit-level communication systems, where resource allocation primarily focuses on bit-level transmission, semantic communication requires strategies that address not only traditional allocation factors (e.g., transmission power and bandwidth) but also semantic-level considerations. In other words, these strategies must ensure the optimal allocation of communication resources to different semantic features or symbols, thereby ensuring the reliability and effectiveness of semantic transmission [25]. In addition, the inherent resource limitations of UAVs restrict their communication range and cause severe signal attenuation or loss [3]. However, the collaborative flight and relay of multiple UAVs can effectively extend coverage, reduce latency, and improve communication reliability. This enables UAVs to transmit semantic information efficiently, even under challenging conditions [26], [27]. Therefore, it is essential to design a high-performance multi-modal semantic communication framework, along with a rational multi-UAV collaborative semantic resource allocation scheme, to establish robust and efficient UAV semantic communication networks.

Several recent studies have focused on resource allocation for semantic communication in UAV networks [27], [28], [29], but significant challenges remain to be addressed. Firstly, it is crucial to evaluate the semantic performance of the UAV semantic communication network, based on which the resource allocation schemes can be evaluated. Since the inherent advantage of semantic communications is to enhance the quality of experience (QoE), it is imperative to define a novel QoE metric that considers both the reduction of the transmission latency and the increased semantic information content. Moreover, due to the energy constraints of UAVs and limited spectrum resources, striking a balance between transmission cost and QoE remains a significant challenge. Secondly, when taking into account multi-modal semantic communication for the UAV, a major challenge lies in the development of a dynamic intelligent transmission architecture. This architecture should enable UAVs to dynamically select and adapt to different modalities based on available resources, ensuring optimal performance in diverse communication scenarios. Thirdly, in multi-UAV semantic communication networks, the collaborative flight and relay among UAVs incur complexities in designing resource allocation schemes. Finally, semantic-aware resource allocation of multi-modal data involves decisions not only on the allocation of communication resources but also on the selection of different semantic symbol numbers. This leads to an optimization problem that involves both discrete and continuous variables, significantly increasing the complexity of the resource allocation process.

Motivated by the aforementioned issues, we extend our prior work in [1] by introducing collaborative relay communication and trajectory optimization and explore resource allocation for multi-modal semantic communication in UAV collaborative networks. We propose a novel intelligent multi-UAV hybrid decision-controlled scheme based on deep reinforcement learning (DRL) to effectively address dynamic optimization problems involving discrete-continuous hybrid variables and multi-UAV collaboration. To the best of the authors' knowledge, it is the first work that addresses resource allocation for multi-modal semantic communication in a UAV collaborative relay network, incorporating a joint optimization of both bit-level and semantic-level resources, as well as UAV trajectories.

The main contributions of this paper are summarized as follows.

- For the multi-modal semantic communication UAV collaborative network, a multi-modal semantic transmission architecture is designed to transmit original image information with different modalities (e.g., image or text). This enables UAVs to adopt adaptive semantic modalities based on the availability of communication resources, achieving efficient transmission. Additionally, a multi-UAV collaborative transmission scheme is implemented, where UAVs serve as relays for one another and coordinate their actions to effectively enhance coverage and improve communication reliability.
- To optimize the resource allocation in UAV communication networks at the semantic level, the QoE metric is defined based on semantic entropy and semantic rate. This metric comprehensively evaluates both transmission latency and the successful reconstruction of semantic information. Furthermore, a semantic-aware resource allocation framework is established to maximize QoE while minimizing the transmission cost (transmission energy and bandwidth) through the joint optimization of UAV trajectories, spectrum bandwidth, transmit power, and the number of the transmitted semantic symbols.
- Due to the presence of discrete and continuous mixed variables, the formulated optimization problem is non-convex and challenging to solve. A multi-UAV hybrid decision-controlled DRL scheme for multi-modal semantic communication resource allocation, named MHDCD-MSC, is proposed to facilitate coordination among different UAVs and enable simultaneous determination of discrete action (the number of semantic symbols) and continuous actions (the transmit power, bandwidth and UAV trajectories) without discretization or relaxation of the action space.

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- Extensive simulations are conducted to demonstrate the effectiveness of the proposed intelligent scheme compared with various benchmark schemes. The simulation results reveal that our proposed scheme can achieve an optimal trade-off between the transmission cost and QoE by taking advantage of the multi-modal semantic communication and an intelligent hybrid decision-controlled scheme based on DRL. Moreover, the UAV collaborative relay communication is shown to significantly enhance the performance of the UAV semantic communication.

The remainder of this paper is organized as follows. Related work is briefly reviewed in Section II. The multi-modal semantic communication UAV collaborative network is presented in Section III. Section IV formulates the semantic-aware optimization problem and proposes a multi-UAV intelligent hybrid decision-controlled DRL scheme. Simulation analysis and evaluation results are presented in Section V. The paper is concluded in Section VI.

II. Related Work

Current works have extensively explored the design of encoding schemes and robust transmission protocols for semantic communication [13], [14], [15], [16], [17], [18]. Preliminary investigations have also been conducted on resource allocation schemes in semantic communication systems [27], [28], [29], [30], [31], [32], [33]. These related works are presented as follows.

A. Semantic Communication System Design

In the field of semantic communication system design, the use of deep learning has been extensively explored, particularly in the domains of text, image, and other related aspects. In [13], a Transformer-based framework for text semantic transmission was introduced with the goal of restoring sentence meaning through semantic information. This approach effectively reduced semantic errors during transmission and enhanced system capacity. Moreover, transfer learning was employed in the training phase to expedite the training of transceivers and facilitate model utilization across diverse communication environments. In the semantic communication for image transmission, the authors in [14] adopted convolutional neural networks (CNNs) to develop a joint source and channel encoding framework. Compared with the traditional encoding methods, this image semantic transfer model has better robustness under challenging channel conditions. Moreover, in addressing challenges associated with multi-modal, multi-user information, the authors in [15] proposed a multi-user semantic communication system that builds corresponding Transformer-based transceivers for different modal tasks. This system constructed distinct Transformer-based transceivers for various modal tasks, with a shared transmitter structure but distinct receiver structures. To reduce the semantic communication processing overhead of different modal tasks, a unified end-to-end semantic communication framework was established in [16]. This framework incorporated a unified feature representation codebook with domain adaptation, as well as a dimension selection module. These advancements enabled the system to adapt to various tasks, resulting in a significant reduction in the transmission overhead and model size. Following this trend, researchers have extended the application of semantic communication for UAV communication lately. The authors in [17] designed a CNN-based semantic communication architecture for UAV image acquisition tasks by introducing temporal information into the traditional CNN structure to improve the efficiency of UAV semantic segmentation and the accuracy of label positioning. To address the limitations imposed by the restricted computing resources of UAVs, a semantic communication model based on the joint perception method was introduced in [18] to extract and compress semantic information for the aerial image transmission scene classification task. Based on the perception of image as well as channel state, this semantic transmission model improved semantic extraction accuracy and reduced the amount of data transmitted by UAVs. These works demonstrated the great potential of semantic communication in reducing network traffic and enhancing communication robustness.

B. Resource Allocation for Semantic Communication

Considering the role of semantic information in resource allocation, several research endeavors have focused on designing semantic-driven resource allocation methods. The authors in [30] proposed a method for compressing semantic information based on task relevance and introduced an adaptive semantic compression ratio for image classification tasks to maximize the success probability of the task by jointly optimizing semantic compression and resource allocation. To evaluate and optimize the efficiency of semantic communication, Kang et al. [31] defined the semantic system throughput and tackled issues concerning resource allocation to optimize the overall semantic system throughput. Nevertheless, it is important to mention that their research primarily concentrated on message-level resource allocation without considering semantic-level aspects. To explore resource allocation in terms of semantics, the authors in [32] defined the transmission rate at the semantic level. They also addressed the problem of maximizing the overall semantic spectrum efficiency for the single-modal text semantic communication. Subsequently, the authors [33] introduced and quantified semantic entropy as a metric for evaluating the amount of multi-modal semantic information. Building upon the concept of semantic entropy, an integrated optimization method was proposed for enhancing semantic communication performance by jointly optimizing semantic symbols, transmit power, and channel allocation. In the works dealing with semantic resource allocation for UAV scenarios, the authors in [27] investigated the uplink semantic communication with the assistance of UAV. They jointly optimized the semantic model scale, resource allocation, and UAV trajectory to improve the efficiency of image data collection in remote areas of the metaverse. The authors in [28] designed a triple-based semantic communication framework for UAV image retrieval tasks. This innovative framework incorporated the personalized priorities to allocate the transmit power to optimize the quality of image transmission. The authors in [29] developed a YOLO-based semantic communication framework

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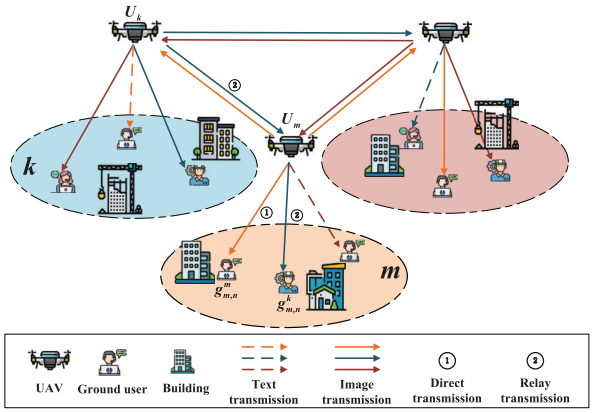


Fig. 1. The multi-modal semantic communication UAV collaborative network.

for UAV image detection. According to the generated confidence, the transmit power was allocated to different parts of the original image to improve the accuracy of the image detection task.

Although significant progress has been made in the design of semantic communication systems [13], [14], [15], [16], [17], [18], and preliminary studies have explored resource allocation in these systems [27], [28], [29], [30], [31], [32], [33], research on resource allocation for semantic communication in UAV networks remains limited. Moreover, several existing studies on UAV semantic communication primarily focus on single-modal tasks [17], [18], [27], [28], [29], leaving the potential of multi-modal approaches largely unexplored. When addressing resource allocation, these works often emphasize traditional bit-level optimization while neglecting the impact of semantic-level factors, e.g., the number of the transmitted semantic symbols, on overall performance [28], [29]. Given that multi-modal semantic communication can enhance the reliability and flexibility, it has the potential to improve the performance of UAV semantic communication networks [15], [16]. Additionally, collaborative relay communication and trajectory optimization have been shown to greatly enhance UAV network performance [26], [27]. In contrast to the existing literature, our study focuses on semantic resource optimization for multi-modal semantic communication in UAV collaborative networks. This research uniquely addresses the joint optimization of UAV trajectory, spectrum bandwidth, transmit power, and the number of the transmitted semantic symbols to improve the overall performance.

III. System Model

The multi-modal semantic communication UAV collaborative network is shown in Fig. 1. We consider a UAV-enabled emergency rescue scenario, where each UAV, assigned to a specific region, captures real-time local image data and transmits the relevant information to ground users within its coverage area. In such urgent and high-risk situations, e.g., large-scale disasters like forest fires or earthquakes, predetermined areas of interest and a known set of ground personnel (e.g., rescue teams, first responders) are commonly established [34]. Following the approach in [35], the entire network is divided into M non-overlapping regions, denoted as $\mathcal{M} = \{1, 2, \ldots, M\}$. In each region $m \in \mathcal{M}$, there is a single UAV and N_m ground users, denoted as U_m and $\mathcal{G}_m = \{g^1_{m,1}, \ldots, g^k_{m,n}, \ldots, g^M_{m,N_m}\}$, where $g^k_{m,n}$ represents the n-th user in region m, who consistently requests real-time information from region $k \in \mathcal{M}$ [36], [37].

Specifically, for the ground user $g^k_{m,n}(m = k)$, requiring real-time information from the local region m, the local U_m UAV establishes direct communication with the ground user to transmit the corresponding semantic information. For the ground user $g^k_{m,n}(m \neq k)$, requiring real-time information from other adjacent regions k, the local U_m UAV acts as a relay UAV, assisting in forwarding the required information from another UAV U_k to the ground user. It is worth noting that during the transmission process, UAVs dynamically adjust between two modalities of transmission (i.e., images or text) to improve service quality and reduce communication resource consumption. The key notations of our system model are listed in Table I for ease of reference.

TABLE I
List of Main Notation

Notation	Description
$m, k, M, \mathcal{M}$	The index, number, and set of regions.
U_m	The UAV of region m.
N_m, $\mathcal{G}_m$	The number, and set of ground users in redion m.
$g^k_{m,n}$	The n-th ground user in region m to receive data from region k.
$t, \mathcal{T}$	The index and set of time slots.
$Z, \vartheta^{U_m}, l^{U_m}$	The flight altitude, angle and distance of UAV U_m.
$\boldsymbol{w}^{U_m}, \boldsymbol{w}_{g^k_{m,n}}$	The horizontal coordinate of UAV U_m and ground user $g^k_{m,n}$.
$PLF^{U_k}_{U_m}$	The A2A path loss between UAV U_k and UAV U_m.
$PL^{U_m}_{g^k_{m,n}}$	The A2G path loss between UAV U_m and ground user $g^k_{m,n}$.
$P^{U_k}_{U_m}, B^{U_k}_{U_m}$	The transmit power and the bandwidth allocated by UAV U_k for UAV U_m.
$P^{U_m}_{g^k_{m,n}}, B^{U_m}_{g^k_{m,n}}$	The transmit power and the bandwidth allocated by UAV U_m for ground user $g^k_{m,n}$.
$\gamma^{U_k}_{U_m}, \gamma^{U_m}_{g^k_{m,n}}$	The SNR between UAV U_k and UAV U_m, and between UAV U_m and user $g^k_{m,n}$.
$\tilde{H}^{U_k}_{U_m}, \tilde{H}^{U_m}_{g^k_{m,n}}$	The approximate semantic entropy between UAV U_k and UAV U_m, and between UAV U_m and user $g^k_{m,n}$.
A^{U_m}, A^{U_k}	The size of the image information collected by UAV UAV U_m and UAV U_k (patchs).
$K^{U_k}_{U_m}, K^{U_m}_{g^k_{m,n}}$	The number of the transmitted semantic symbols.
$R^{U_k}_{U_m}, R^{U_m}_{g^k_{m,n}}$	The semantic rate between UAV U_k and UAV U_m, and between UAV U_m and user $g^k_{m,n}$.
$D^{U_k}_{U_m}, D^{U_m}_{g^k_{m,n}}$	The transmission time between UAV U_k and UAV U_m, and between UAV U_m and user $g^k_{m,n}$.
$D_{g^k_{m,n}}$	The total transmission time of ground user $g^k_{m,n}$.
$QoE_{g^k_{m,n}}$	The QoE of ground user $g^k_{m,n}$.
P^{U_m}, B^{U_m}	The total transmit power and bandwidth consumed by UAV U_m.
C^{U_m}	The transmission cost of UAV U_m.

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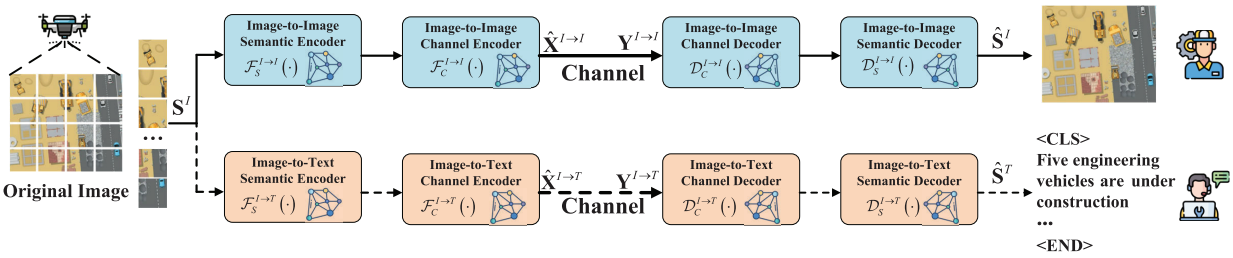


Fig. 2. The framework for the multi-modal semantic communication.

A. Multi-Modal Semantic Communication

As shown in Fig. 2, the multi-modal semantic communication model consists of both the image-to-image and image-to-text semantic transceivers. The former extracts image-to-image semantic features at the transmitter to accurately transmit the image to the user. In contrast, the latter extracts image-to-text semantic features at the transmitter and then transforms the essential information of the original image into a textual description for the user, which greatly reduces data transmission volume and ensures relatively effective communication even when communication resources are extremely limited. The UAV can dynamically select one of these two modalities based on the availability of communication resources to achieve effective transmission.

The image-to-image and image-to-text transmitters have similar network structures, where their semantic encoders employ multiple vision Transformer layers [38], [39], and channel encoders utilize multiple dense layers to generate transmission symbols. The image captured by the UAV is segmented and reshaped into a sequence of patches $\mathbf{S}^I = \{p_1, p_2, \ldots, p_A\}$ as inputs, where $I \subseteq \{I : image,\ T : text\}$ indicates the modality of the original information, A denotes the total number of patches. Through semantic extraction and compression of the original image information $\mathbf{S}^I$, the transmitter of each semantic transceiver produces the corresponding semantic symbol vector $\mathbf{X}^{I\to J} \in \Re^{1\times A\times K_J}$ for channel transmission, given as

$$\mathbf{X}^{I\to J} = \mathcal{F}_C^{I\to J}\left(\mathcal{F}_S^{I\to J}\left(\mathbf{S}^I\left|\boldsymbol{\sigma}_J\right.\right)\left|\boldsymbol{\varsigma}_J\right.\right), \tag{1}$$

where $J \subseteq \{I : image,\ T : text\}$ is the modality of the information received by the ground user, K_J denotes the average number of the mapped semantic symbols for each patch when the original image is transmitted with modality J. $\mathcal{F}_S^{I\to J}\left(\cdot|\boldsymbol{\sigma}_J\right)$ represents the semantic encoder with the parameter set $\boldsymbol{\sigma}_J$, and $\mathcal{F}_C^{I\to J}\left(\cdot|\boldsymbol{\sigma}_J\right)$ represents the channel encoder with the parameter set $\boldsymbol{\varsigma}_J$. The semantic symbols are transmitted over a wireless channel, and ground users receive a signal $\mathbf{Y}^{I\to J} = h\mathbf{X}^{I\to J} + \mathbf{n}$, where h represents the channel gain and $\mathbf{n}$ is the additive white Gaussian noise (AWGN).

The receiver of each ground user consists of a channel decoder and a semantic decoder. The channel decoder employs multiple dense layers with varying units, while multiple Transformer decoder layers are utilized for the semantic decoder. Those two decoders work together to decode the received signal and output the information with J modality $\hat{\mathbf{S}}^J$, given as

$$\hat{\mathbf{S}}^J = \mathcal{D}_S^{I\to J}\left(\mathcal{D}_C^{I\to J}\left(\mathbf{Y}^{I\to J}\left|\boldsymbol{\psi}_J\right.\right)\left|\boldsymbol{\xi}_J\right.\right), \tag{2}$$

where $\mathcal{D}_C^{I\to J}\left(\cdot|\boldsymbol{\psi}_J\right)$ and $\mathcal{D}_S^{I\to J}\left(\cdot|\boldsymbol{\xi}_J\right)$ respectively represent the channel decoder and the semantic decoder. $\boldsymbol{\psi}_J$ and $\boldsymbol{\xi}_J$ are their respective parameter sets.

Moreover, the accuracy of each semantic transceiver is reflected from the difference between the target information $\mathbf{S}^J$ and the estimated information $\hat{\mathbf{S}}^J$, expressed as $\eta_J(\mathbf{S}^J, \hat{\mathbf{S}}^J)$, where the image-to-image and image-to-text transceivers are evaluated by using the structural similarity (SSIM) and bilingual evaluation understudy (BLEU) score, respectively [16].

According to [32], the accuracy metric $\eta_J(\mathbf{S}^J, \hat{\mathbf{S}}^J)$ primarily depends on two factors, i.e., the signal-to-noise ratio (SNR) γ of the channel and the number of the transmitted semantic symbols for each patch (i.e., K_J). Therefore, the accuracy metrics can be expressed as functions of K_J and γ, i.e., $\eta_J(\mathbf{S}^J, \hat{\mathbf{S}}^J) = f_J\left(K_J, \gamma\right)$. However, there is a lack of an explicit form for the accuracy function $f_J\left(K_J, \gamma\right)$. To overcome this obstacle, the curve fitting method is employed to approximate the accuracy function [40], where the sigmoid function serves as the function model for fitting, expressed as

$$f_J\left(K_J, \gamma\right) \approx \frac{L_{K_J}}{1 + \exp\{-\zeta_{K_J}(\gamma - \gamma_{K_J})\}} + b_{K_J} \tag{3}$$

where L_{K_J} and b_{K_J} denote upper (right) asymptote and lower (left) asymptote, respectively. ζ_{K_J} denotes the growth rate, γ_{K_J} controls the mid-point. The final fitted function curves are shown in Fig. 4 and 5 of Section IV.

1) Approximate Semantic Entropy and Semantic Rate: Since the amount of semantic information depends on the source data and the specific task [32], the measurement of semantic information needs to be defined according to the particular situation. To this end, semantic entropy [33] was proposed to measure the semantic information for a given source and task. It is characterized as the minimum number of semantic symbols required for each patch to meet the accuracy requirement without considering the channel effect. Following [33], the approximate semantic entropy can be expressed as

$$\begin{aligned}\tilde{H}_J\left(S^I; \hat{S}^J\right) \triangleq \min \mathbb{E}\left(\dim\left(Code^{E_{I\to J}}\left(S\right); 3\right)\right)\\ s.t.\ \eta_J\left(S^J, \hat{S}^J\right) > \nu,\end{aligned} \tag{4}$$

where $\tilde{H}_J$ represents the approximate semantic entropy when the estimated information $\hat{S}^J$ is obtained based on the source data S^I, $Code^{E_{I\to J}}\left(S\right)$ represents the semantic symbol vector

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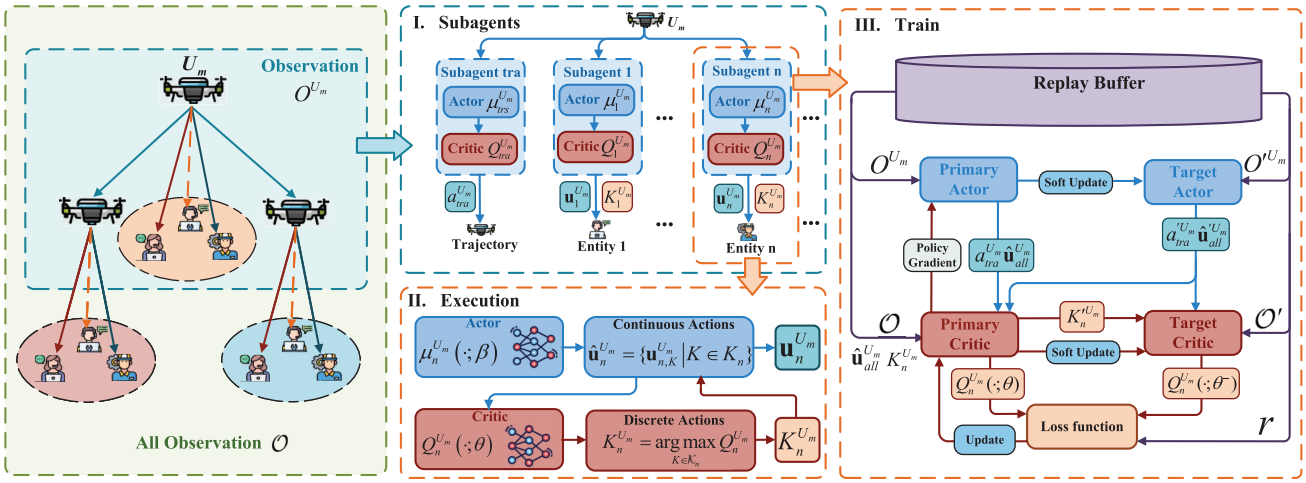


Fig. 3. The framework of our proposed MHDCD-MSC scheme.

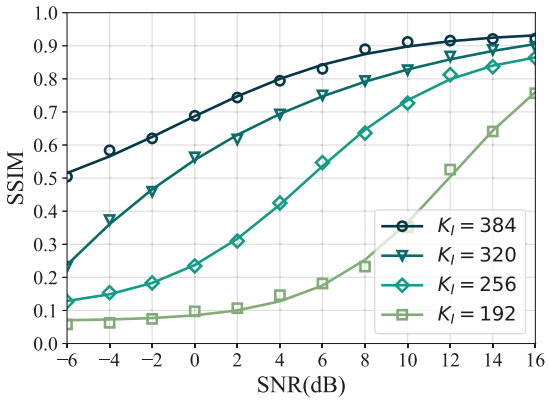


Fig. 4. SIMM versus SNR and associated fitting functions for image-to-image transceivers.

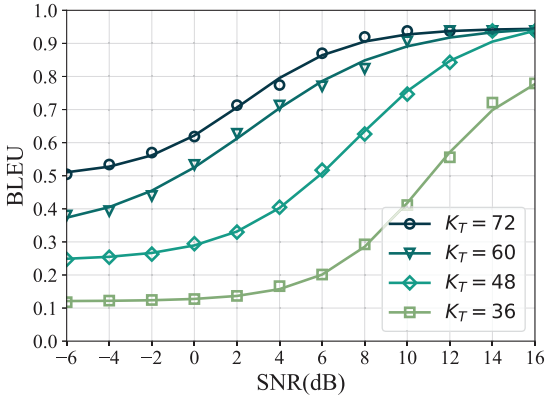


Fig. 5. BLEU score versus SNR and associated fitting functions for image-to-text transceivers.

$\mathbf{X}^{I\to J} \in \Re^{1\times A\times K_J}$ extracted by the joint semantic-channel encoder $E_{I\to J}$ from S^I, and $\dim(\cdot;3)$ represents the size of the third dimension of the semantic symbol vector, i.e., the number of semantic symbols for each patch, K_J. The constraint ensures that the semantic encoder can accurately transmit the target information, and ν denotes the minimum requirement for the semantic accuracy.

If the transmission modality J is fixed, the approximate semantic entropy $\tilde{H}_J$ is a constant. The unit of the approximate semantic entropy is defined as '$sut/patch$', which represents the average semantic information contained by each patch of the image. The semantic information per semantic symbol is given by $\tilde{H}_J/K_J$ $(suts/symbol)$. Recall the fact that the symbol rate is equal to the transmission bandwidth, which is denoted by B. Hence, the semantic rate $(suts/s)$ can be represented as [33]

$$R = B\frac{\tilde{H}_J}{K_J} f_J(K_J,\gamma). \tag{5}$$

B. Multi-UAV Collaborative Transmission Model

In the considered multi-modal semantic communication UAV collaborative network, the transmission model is assumed to work in a time-slotted manner, denoted as $\mathcal{T} = \{1,2,\ldots,t,\ldots\}$, where each time slot t has the duration of τ. All UAVs move at a fixed altitude Z [41]. The horizontal coordinate of UAV U_m at time t is denoted by $\boldsymbol{w}^{U_m}(t) = \left(x^{U_m}(t), y^{U_m}(t)\right)$. At the same time, all ground users move randomly within the specific area. The horizontal coordinate of ground user $g^k_{m,n}$ at time t is denoted by $\boldsymbol{w}_{g^k_{m,n}}(t) = (x_{g^k_{m,n}}(t), y_{g^k_{m,n}}(t))$. Assume that UAV U_m flies the distance $l^{U_m}(t)$ with the angle direction $\vartheta^{U_m}(t) \in [0, 2\pi)$ in the horizontal flight. Then, the coordinate of UAV U_m at next time slot $t+1$ is expressed as

$$x^{U_m}(t+1) = x^{U_m}(t) + l^{U_m}(t)\cos(\vartheta^{U_m}(t)), \tag{6a}$$
$$y^{U_m}(t+1) = y^{U_m}(t) + l^{U_m}(t)\sin(\vartheta^{U_m}(t)). \tag{6b}$$

Due to the limited horizontal-flight speeds of UAVs, they always have limited flight distances, which can be given by

$$0 \le l^{U_m}(t) \le l_{\max}, \tag{7}$$

where $l_{\max}$ represents the maximum horizontal flight distance of UAVs. Moreover, in order to guarantee that UAVs move

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within the specific service area, the following move constraint must be satisfied [42], that is,

$$0 \leq \left\| \boldsymbol{w}^{U_m}(t) - \boldsymbol{w}^{U_m}(0) \right\| \leq w_{\max}, \tag{8}$$

where $\boldsymbol{w}^{U_m}(0)$ is the initial position of the UAV U_m, located in the center of the region m. $w_{\max}$ represents the radius of the region m.

During the whole transmission, if the ground user $g_{m,n}^k$ intends to receive the real-time image information from its local region, i.e., $m = k$, its associated UAV U_m transfers the corresponding information to user $g_{m,n}^k$ directly. In the other case, when user $g_{m,n}^k$ aims to require the image information from other adjacent regions, i.e., $m \neq k$, due to the long distance between the source UAV U_k and the ground user $g_{m,n}^k$, extra energy consumption is incurred. Thus, a UAV collaborative relay scheme is explored, i.e., the image information is first relayed to UAV U_m from UAV U_k, and then user $g_{m,n}^k$ can finally acquire the image information of region k. Thus, two-channel models are considered for UAV relaying mode, i.e., the air-to-air (A2A) transmission model between UAVs and the air-to-ground (A2G) downlink transmission model between UAVs and the ground users.

1) A2A Transmission Model: The communication link between UAVs is assumed to be a Line-of-Sight (LoS) link due to the negligible blocking in the air [43]. The A2A path loss between UAV U_k and UAV U_m follows the free space path loss model [44], denoted as

$$PLF_{U_m}^{U_k}(t) = 20\log d_{U_m}^{U_k}(t) + 20\log f_c + 20\log\left(\frac{4\pi}{v_c}\right), \tag{9}$$

where $d_{U_m}^{U_k}(t) = \|\boldsymbol{w}_{U_m}(t) - \boldsymbol{w}_{U_k}(t)\|$ is the distance between UAV U_m and UAV U_k. f_c denotes the carrier frequency, and v_c represents the velocity of light. The UAVs adopt the frequency division multiple access (FDMA) mode to serve ground users and other UAVs. Then, the channel SNR at UAV U_m from UAV U_k is expressed as

$$\gamma_{U_m}^{U_k}(t) = \frac{P_{U_m}^{U_k}(t) 10^{-PLF_{U_m}^{U_k}(t)/10}}{B_{U_m}^{U_k}(t) n_0}, \tag{10}$$

where $P_{U_m}^{U_k}(t)$ and $B_{U_m}^{U_k}(t)$ represent the transmit power and the bandwidth allocated by UAV U_m for A2A transmission with UAV U_k, respectively, and n_0 is the noise power spectral density.

Considering the favorable communication environment of the A2A transmission model [43], to ensure the overall communication quality of the network while simplifying the communication process, UAVs exclusively communicate with each other in image modality. Thus, the UAV U_k adopts the image-to-image transceiver to send information with image modality, and the chosen number of the transmitted semantic symbols is $K_{U_m}^{U_k}(t) \in \mathcal{K}_I$, and $\mathcal{K}_I = \{K_I^1, K_I^2, \ldots\}$ is the set of the available transmitted semantic symbol numbers for the image-to-image semantic communication. Correspondingly, the approximate semantic entropy $\tilde{H}_{U_m}^{U_k}(t)$ and the accuracy function $f_{U_m}^{U_k}(\cdot)$ are respectively equal to $\tilde{H}_I$ and $f_I(\cdot)$. According to the definitions of semantic entropy and semantic rate in Sec. III-A, the achievable semantic rate between UAV U_k and UAV U_m at time slot t is

$$R_{U_m}^{U_k}(t) = \frac{\tilde{H}_I(t)}{K_{U_m}^{U_k}(t)/B_{U_m}^{U_k}(t)} f_I\left(K_{U_m}^{U_k}(t), \gamma_{U_m}^{U_k}(t)\right). \tag{11}$$

Let the size of the image information collected by UAV U_k be $A^{U_k}(t)$ patches. The total amount of semantic information sent by UAV U_k to the UAV U_m is $A^{U_k}(t)\tilde{H}_{U_m}^{U_k}(t)$, and the transmission time between UAV U_k and UAV U_m is formulated as

$$D_{U_m}^{U_k}(t) = \frac{A^{U_k}(t)\tilde{H}_I(t)}{R_{U_m}^{U_k}(t)}. \tag{12}$$

2) A2G Transmission Model: In the multi-modal semantic communication UAV collaborative network, UAVs establish direct connections with the local ground users via the A2G channel. Due to the occlusion effect caused by structures in the practical scenarios, UAVs connect to the ground users through LoS or Non-Line-of-Sight (NLoS) links with different probabilities [44]. According to the probabilistic pathloss model in [45], the probability of LoS link between the UAV U_m and user $g_{m,n}^k$ at time slot t is

$$PLOS_{g_{m,n}^k}^{U_m}(t) = \frac{1}{1 + \eta_a \exp\left\{-\eta_b\left[\arcsin\left(\frac{Z}{d_{g_{m,n}^k}^{U_m}(t)}\right) - \eta_a\right]\right\}}, \tag{13}$$

where η_a and η_b are constants related to the type of the propagation environment. $d_{g_{m,n}^k}^{U_m}(t) = \sqrt{\left\|\boldsymbol{w}_{g_{m,n}^k}(t) - \boldsymbol{w}_{U_m}(t)\right\|^2 + Z^2}$ is the distance between UAV U_m and ground user $g_{m,n}^k$. Therefore, the probability of the NLoS link is given as

$$PNLOS_{g_{m,n}^k}^{U_m}(t) = 1 - PLOS_{g_{m,n}^k}^{U_m}(t). \tag{14}$$

Similar to [44], the A2G path loss between UAV U_m and user $g_{m,n}^k$ is modeled as

$$PL_{g_{m,n}^k}^{U_m}(t) = PLF_{g_{m,n}^k}^{U_m}(t) + PLOS_{g_{m,n}^k}^{U_m}(t) \times \varphi_{LOS} + PNLOS_{g_{m,n}^k}^{U_m}(t) \times \varphi_{NLOS}, \tag{15}$$

where $PLF_{g_{m,n}^k}^{U_m}(t)$ is the free space pathloss defined in eq. (9), φ_{LOS} and φ_{NLOS} denote the excessive pathloss for LoS and NLoS links, respectively. Let $B_{g_{m,n}^k}^{U_m}(t)$ be the bandwidth allocated to user $g_{m,n}^k$.

Moreover, the small-scale fading of the A2G channel can be modeled following the approach in [46]. Specifically, for Line-of-Sight (LoS) links, the small-scale fading $o_{g_{m,n}^k}^{U_m}(t)$ between the UAV U_m and user $g_{m,n}^k$ can be modeled as Rician fading, expressed as

$$o_{g_{m,n}^k}^{U_m}(t) = \sqrt{\frac{V_{g_{m,n}^k}^{U_m}(t)}{V_{g_{m,n}^k}^{U_m}(t)+1}}\, o + \sqrt{\frac{V_{g_{m,n}^k}^{U_m}(t)}{V_{g_{m,n}^k}^{U_m}(t)+1}}\, \tilde{o}, \tag{16}$$

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where o represents a deterministic LoS channel component with an absolute value of 1, and $\tilde{o}$ is a zero-mean unit variance circularly symmetric complex Gaussian (CSCG) random variable. $V_{g_{m,n}^k}^{U_m}(t)$ represents the Rician factor of the channel between the UAV U_m and user $g_{m,n}^k$, expressed as

$$V_{g_{m,n}^k}^{U_m}(t) = v_1 \exp\left\{ v_2 \arcsin\left(\frac{Z}{d_{g_{m,n}^k}^{U_m}(t)} \right) \right\}, \tag{17}$$

where v_1 and v_2 are constant coefficients determined according to the specific environment. For NLoS links, small-scale fading $o_{g_{m,n}^k}^{U_m}(t)$ between the UAV U_m and user $g_{m,n}^k$ can be modeled as Rayleigh fading, which is equivalent to the Rician fading model where $V_{g_{m,n}^k}^{U_m}(t) = 0$ [46]. Furthermore, for other types of signal attenuation, such as those caused by Doppler effects, we assume that similar approaches of [46], and [47] can be effectively applied at the receiver to achieve adequate compensation.

Then, the SNR between UAV U_m and ground user $g_{m,n}^k$ is expressed as

$$\gamma_{g_{m,n}^k}^{U_m}(t) = \frac{P_{g_{m,n}^k}^{U_m}(t)\, h_{g_{m,n}^k}^{U_m}(t)}{B_{g_{m,n}^k}^{U_m}(t)\, n_0}, \tag{18}$$

where $h_{g_{m,n}^k}^{U_m}(t) = \left| o_{g_{m,n}^k}^{U_m}(t) \right|^2 10^{-PL_{g_{m,n}^k}^{U_m}(t)/10}$ is the channel gain between UAV U_m and ground user $g_{m,n}^k$, $P_{g_{m,n}^k}^{U_m}(t)$ and $B_{g_{m,n}^k}^{U_m}(t)$ represent the transmit power and bandwidth allocated by UAV U_m for A2G transmission with the ground user $g_{m,n}^k$, respectively.

When UAV U_m serves as a relay to assist UAV U_k for transmitting information to the ground user $g_{m,n}^k$, $m \neq k$, UAV U_m further performs semantic modality selection on the information received from UAV U_k to select an appropriate semantic modality (image or text) based on channel conditions to transmit the information to the ground user $g_{m,n}^k$.

Then, the achievable semantic rate between UAV U_m and the ground user $g_{m,n}^k$ at time slot t is given as

$$R_{g_{m,n}^k}^{U_m}(t) = \frac{\tilde{H}_{g_{m,n}^k}^{U_m}(t)}{K_{g_{m,n}^k}^{U_m}(t) / B_{g_{m,n}^k}^{U_m}(t)} f_{g_{m,n}^k}^{U_m}\left(K_{g_{m,n}^k}^{U_m}(t), \gamma_{g_{m,n}^k}^{U_m}(t)\right), \tag{19}$$

where $K_{g_{m,n}^k}^{U_m}(t) \in \mathcal{K}$ represents the number of the transmitted semantic symbols chosen by UAV U_m for the ground user $g_{m,n}^k$. The set $\mathcal{K} = \mathcal{K}_T \cup \mathcal{K}_I$, $\mathcal{K}_T \cap \mathcal{K}_I = \emptyset$ encompasses all available transmitted semantic symbol numbers for both image-to-text and image-to-image semantic communication. The modality selection of UAV U_m is determined by the chosen number of the transmitted semantic symbols $K_{g_{m,n}^k}^{U_m}$. If $K_{g_{m,n}^k}^{U_m}(t) \in \mathcal{K}_T$, it means that the UAV U_m adopts the image-to-text transceiver to send information with text modality. In this scenario, the approximate semantic entropy is given by $\tilde{H}_{U_m}^{U_k}(t) = \tilde{H}_T$, and the accuracy function is represented as $f_{g_{m,n}^k}^{U_m}(\cdot) = f_T(\cdot)$.

Then, the transmission time between UAV U_m and ground user $g_{m,n}^k$ is formulated as

$$D_{g_{m,n}^k}^{U_m}(t) = \frac{A^{U_k}(t)\, \tilde{H}_{g_{m,n}^k}^{U_m}(t)}{R_{g_{m,n}^k}^{U_m}(t)}. \tag{20}$$

Consequently, the total transmission time spent for ground user $g_{m,n}^k$ to acquire target information of the region k can be expressed as

$$D_{g_{m,n}^k}(t) = \begin{cases} D_{g_{m,n}^k}^{U_m}(t), & k = m; \\ D_{U_m}^{U_k}(t) + D_{g_{m,n}^k}^{U_m}(t), & k \neq m, \end{cases} \tag{21}$$

where $k = m$ means that the ground user $g_{m,n}^k$ aims to receive information originating from the local region. In this case, UAV U_m uses direct communication to transmit semantic information to the ground user $g_{m,n}^k$, and the total transmission time only consists of the A2G transmission time $D_{g_{m,n}^k}^{U_m}(t)$. On the other hand, $k \neq m$ means that ground user $g_{m,n}^k$ and the source region k of the target image belong to two different regions. In this scenario, UAV U_k and UAV U_m employ single-hop relay communication to transmit information to the ground user $g_{m,n}^k$. The time duration of UAV relay nodes to process information can be negligible compared to the transmission time [7]. Therefore, the total transmission time is composed of A2A transmission time $D_{U_m}^{U_k}(t)$ and A2G transmission time $D_{g_{m,n}^k}^{U_m}(t)$.

C. UAV Quality of Experience and Transmission Cost

In the considered multi-modal semantic communication UAV collaborative network, the requirement of users is to receive as much semantic information as possible while while ensuring a certain timeliness of the received information. To comprehensively evaluate the quality of user experience in the UAV communication network, we define a QoE function that incorporates transmission time and the received semantic information amount, expressed as

$$QoE_{g_{m,n}^k}(t) = \frac{1}{1 + \exp\{\zeta_D (D_{g_{m,n}^k}(t) - D_{th})\}} \times \frac{1}{1 + \exp\{\zeta_H A^{U_k}(t)(H_{th} - H_{g_{m,n}^k}^{U_m}(t))\}}, \tag{22}$$

where these two terms on the right side of eq. (22) represent the scores for the transmission time and the received semantic information amount of user $g_{m,n}^k$, respectively. These two terms are scored by the sigmoid function and multiplied together, providing a comprehensive and sensitive method to describe their correlation [48]. $D_{g_{m,n}^k}(t)$ is the total time consumed by the ground user $g_{m,n}^k$ to receive information of the region k. $A^{U_k}(t)$ represents the number of patches transmitted from UAV U_k to user $g_{m,n}^k$. $\tilde{H}_{g_{m,n}^k}^{U_m}(t)$ denotes the corresponding semantic entropy of the information received by the target user $g_{m,n}^k$. ζ_D and ζ_H are the growth rates of the transmission time score and the information amount score. D_{th} is the maximum transmission time and H_{th} is the minimum semantic entropy.

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Since UAVs have inherently limited resources, we consider the transmission cost[1] in our model, which is composed of the transmit power cost and the bandwidth cost of UAVs, expressed as

$$C^{U_m}(t) = \alpha_p \frac{P^{U_m}(t)}{P_{\max}} + \alpha_b \frac{B^{U_m}(t)}{B_{\max}}, \tag{23}$$

where $P_{\max}$ represents the maximum available transmit power, and $B_{\max}$ is the maximum available bandwidth. α_p and α_b are the weighted parameters. The total transmit power $P^{U_m}(t)$ and bandwidth $B^{U_m}(t)$ consumed by UAV U_m are expressed as, respectively,

$$P^{U_m}(t) = \sum_{n=1}^{N_m} P^{U_m}_{g^k_{m,n}}(t) + \sum_{k=1,k\neq m}^{M} P^{U_m}_{U_k}(t), \tag{24a}$$

$$B^{U_m}(t) = \underbrace{\sum_{n=1}^{N_m} B^{U_m}_{g^k_{m,n}}(t)}_{A2G} + \underbrace{\sum_{k=1,k\neq m}^{M} B^{U_m}_{U_k}(t)}_{A2A}, \tag{24b}$$

where the first parts of Eq. 24a and Eq. 24b denote the transmit power and bandwidth consumed by the UAV U_m for A2G communication, while the latter parts indicate the transmit power cost and the bandwidth cost of the UAV U_m during the A2A communication.

IV. Intelligent Resource Allocation for Multi-Modal Semantic Communication UAV Collaborative Networks

A. Problem Formulation

In this section, a semantic-aware resource allocation problem is formulated to maximize the QoE while minimizing the communication resource consumption in the long term by allocating the number of the UAV trajectories, the transmitted semantic symbols, the transmit power, and the spectrum bandwidth. The QoE-cost tradeoff problem is formulated as

$$\max_{\{l,\theta,K,P,B\}} \ \frac{1}{|T|}\sum_{t\in T}\sum_{m\in M}(\alpha_q Q_o E_m(t) - \alpha_c C^U_m(t)) \tag{25}$$

$$\text{s.t. } C_1: \ 0 \le l_m(t) \le l_{\max}, \tag{25a}$$
$$C_2: \ \vartheta_m(t) \in [0, 2\pi), \tag{25b}$$
$$C_3: \ 0 \le \|w_{U_m}(t) - w_{U_m}(0)\| \le w_{\max}, \tag{25c}$$
$$C_4: \ K^{U_k}_{U_m}(t) \in \mathcal{K}_{\mathcal{I}}, K^{U_m}_{g^k_{m,n}}(t) \in \mathcal{K}, \tag{25d}$$
$$C_5: \ 0 \le P^{U_m}(t) \le P_{\max}, \tag{25e}$$
$$C_6: \ 0 \le B^{U_m}(t) \le B_{\max}, \tag{25f}$$
$$C_7: \ 0 \le D_{g^k_{m,n}}(t) \le D_{th}, \tag{25g}$$
$$C_8: \ \phi^{U_m}_{g^k_{m,n}}, \phi^{U_k}_{U_m} \ge \phi_{th}, \tag{25h}$$

where $QoE_m(t) = \sum_{k=1}^{M}\sum_{n=1}^{N_m} QoE_{g^k_{m,n}}(t)$ represents the sum of the QoE for all users in the region m. α_q and α_c are the weighted parameters to achieve a tradeoff between the QoE and transmission cost. The objective function (25) is to maximize the QoE of the ground users in each region while minimizing the transmission cost of UAVs. C_1, C_2 and C_3 are constraints of the UAV trajectories. C_4 specifies the optional number of the transmitted semantic symbols. C_5 and C_6 are the range of the transmit power and bandwidth. C_7 and C_8 guarantee the transmission time and accuracy requirements. $\phi^{U_m}_{g^k_{m,n}}$ and $\phi^{U_k}_{U_m}$ are derived from the accuracy functions $f^{U_m}_{g^k_{m,n}}(\cdot)$ and $f^{U_k}_{U_m}(\cdot)$, representing the semantic transmission accuracies for A2G and A2A communication, respectively. ϕ_{th} represents the threshold of the transmission accuracy.

Due to the non-convex objective function and constraints, the problem (25) is highly non-convex and challenging to solve. Moreover, the discrete choice of semantic symbols and the continuous decisions regarding the UAV trajectories and communication resource allocation are coupled with each other, which further complicates the joint decision-making process. As a result, seeking real-time optimal solutions for the problem using conventional optimization methods in a dynamic environment is complex and unfeasible. In the following, we adopt DRL to address these challenges. Specifically, we propose a hybrid decision-controlled scheme based on DRL, where each UAV is modeled as an agent composed of multiple subagents. Each subagent collaboratively learns to make decisions on the selection of semantic symbols, the UAV trajectories and the allocation decisions for communication resources. In the subsequent section, we reformulate this problem into a Markov Decision Process (MDP) and introduce a multi-UAV hybrid decision-controlled DRL scheme.

B. MDP Modeling

To model MDP for the optimization problem, the multi-modal semantic communication UAV collaborative network is treated as the environment, with the UAVs acting as agents. These three key elements of the UAV agent U_m are defined as follows.

1) Observation Space: The observation space of UAV U_m can be formulated as

$$O^{U_m}(t) = \{A^{U_m}(t), \mathcal{W}^{U_m}(t), C^{U_m}(t-1), \mathcal{R}^{U_m}(t-1)\}, \tag{26}$$

where $A^{U_m}(t)$ represents the size of image information collected by UAV U_m from the region m. $\mathcal{W}^{U_m}(t) = \{\boldsymbol{w}^{U_m}(t-1), \boldsymbol{w}_g(t)\}_{g\in\mathcal{G}_m}$ denotes the location of UAV U_m and ground users in region m. $C^{U_m}(t-1)$ refers to the transmission cost of UAV U_m in the previous time slot. $\mathcal{R}^{U_m}(t-1) = \{\gamma^{U_m}_g, \gamma^{U_m}_{U_k}\}_{g\in\mathcal{G}_m, k\in\mathcal{M}\setminus\{m\}}$ represents the SNR of the channels between UAV U_m and its directly associated entities, which include local ground users and other UAVs.

2) Action Space: The action space of UAV U_m encompasses the actions taken by U_m for each ground user or UAV with which it directly communicates. Formally, the action space is characterized as $a^{U_m}(t) = \{l^{U_m}(t), \vartheta^{U_m}(t), K^{U_m}_n(t), P^{U_m}_n(t), B^{U_m}_n(t)\}_{n\in\mathcal{N}_m}$, where $\mathcal{N}_m = \{g, U_k\}_{g\in\mathcal{G}_m, k\in\mathcal{M}\setminus\{m\}}$ denotes the set of

[1]It should be noted that while the mobility power cost of UAVs is non-negligible in absolute terms, it remains relatively constant or varies slowly compared to the rapidly fluctuating transmit power cost in short-duration emergency tasks. Thus, similar to [27], [35], [41], and [42], we focus solely on the transmit cost and disregard the UAV mobility power cost.

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entities directly associated with UAV U_m. [2] $l^{U_m}(t)$ and $\vartheta^{U_m}(t)$ represent the flight distance and angle of UAV U_m, respectively $K_n^{U_m}(t)$ represents the number of semantic transmitted symbols chosen by UAV U_m when directly communicates with entity n. If $K_n^{U_m}(t) \in \mathcal{K}_T$, it means that the UAV U_m adopts text modality to transmit information to entity n. Otherwise, the UAV U_m adopts image modality to transmit information to entity n. $P_n^{U_m}(t)$ and $B_n^{U_m}(t)$ denote the transmit power and bandwidth allocated by UAV U_m for entity n.

3) Reward: In the MDP, the reward function serves to measure the impact of actions taken by the agent at a given state. Considering the common objectives of the formulated optimization problem, all UAV agents should collaboratively work to maximize the QoE of ground users, while minimizing the transmission costs for the UAVs. Moreover, the reward function should be consistent with the system's objective function and constraints. Thus, the reward is composed of three parts. The first two parts are the reward of QoE and the normalized transmission cost, while the third part introduces the penalty for instances where the semantic transmission accuracy does not meet the desired criteria. The penalty for not meeting the minimum semantic accuracy requirement is defined as

$$\rho_n^{U_m}(t) = \begin{cases} 1, & \phi_n^{U_m} \geq \phi_{th}; \\ -1, & \text{otherwise}, \end{cases} \tag{27}$$

where $n \in \mathcal{N}_m$ denotes the ground users or UAVs that directly communicate with UAV m. Then, the reward function can be expressed as

$$r(t) = \sum_{m=1}^{M} \left(\alpha_q QoE_m(t) - \alpha_c C^{U_m}(t) + \xi \rho^{U_m}(t) \right), \tag{28}$$

where $\rho^{U_m}(t)$ denotes the total penalty incurred by UAV U_m, and ξ represents the weight coefficient of the penalty.

C. Multi-UAV Hybrid Decision Controlled DRL Based Multi-Modal Semantic Communication Resource Allocation

Since the UAV U_m is required to optimize its trajectory while providing communication services to multiple ground users and other UAVs, it faces the complex challenge of managing numerous actions concurrently, which significantly increases computational complexity. To address this, we partition the action space of UAV U_m into multiple components and employ multiple subagents within UAV U_m to handle these components in a distributed manner. Specifically, the action space of UAV U_m is initially divided into two main parts. The first part is the trajectory action space $a_{tra}^{U_m} = \{l^{U_m}, \vartheta^{U_m}\}$, and the second part is the semantic resource allocation action space $a_{scr}^{U_m} = \{K_n^{U_m}, P_n^{U_m}, B_n^{U_m}\}_{n \in \mathcal{N}_m}$. Upon further examination of the semantic resource allocation action space, it is observed that $\{K_n^{U_m}, P_n^{U_m}, B_n^{U_m}\}$ at the same entity n always exhibit significant correlations. In response to this, we further partition the semantic resource allocation action space as $a_{scr}^{U_m} = \{a_n^{U_m}\}_{n \in \mathcal{N}_m}$ where $a_n^{U_m} = \{K_n^{U_m}, P_n^{U_m}, B_n^{U_m}\}$ represents the semantic resource allocation action space for entity n.

With the above action space separation, we utilize the actor-critic architecture to develop a multi-UAV hybrid decision-controlled DRL scheme for multi-modal semantic resource allocation, named MHDCD-MSC. The framework of the MHDCD-MSC scheme is illustrated in Fig. 3. Taking the UAV U_m as an example, illustrated in part I of Fig. 3, for the UAV trajectory, the trajectory optimization subagent utilizes the Deep Deterministic Policy Gradient (DDPG) network [50] to generate the UAV's flight distance and flight angle, represented as

$$a_{tra}^{U_m} = \{l^{U_m}, \vartheta^{U_m}\} = \mu_{tra}^{U_m}\left(O^{U_m}; \beta\right), \tag{29}$$

where $\mu_{tra}^{U_m}(;\beta)$represents the actor network responsible for generating trajectory actions for UAV U_m, and β denotes the set of parameters of the actor network.

For semantic resource allocation, the action space $a_n^{U_m}$ for entity n consists of both discrete and continuous components, which cannot be easily solved using typical reinforcement learning models. We develop a hybrid decision-controlled DRL scheme for the subagent managing entity n. The semantic resource allocation action space $a_n^{U_m}$ is divided into two parts, namely, the discrete part for selecting the number of semantic symbols $K_n^{U_m}$, and the continuous part $\mathbf{u}_n^{U_m} = (P_n^{U_m}, B_n^{U_m})$ comprising the transmit power and bandwidth. Based on the selected number of semantic symbols, the optimal transmission power and bandwidth for each entity are first determined. Subsequently, the optimal choice for the number of semantic symbols is refined.

Specifically, as shown in part II of Fig. 3, based on the deterministic policy network, the subagent of UAV U_m for entity n utilizes the actor to explore all the discrete actions and generate the corresponding continuous actions for entity n, expressed as

$$\hat{\mathbf{u}}_n^{U_m} = \mu_n^{U_m}\left(O^{U_m}; \beta\right) = \{\mathbf{u}_{n,K}^{U_m} | K \in \mathcal{K}_n\}, \tag{30}$$

where $\mu_n^{U_m}(;\beta)$ denotes the actor network employed to generate actions for entity n, β is the set of parameters of the actor network. $\mathcal{K}_n$ is the set of all possible discrete actions for entity n. $\mathbf{u}_{n,K}^{U_m}$ is the optimal continuous action when the number of the transmitted semantic symbols $K_n^{U_m} = K$.

Then, the critic network takes the observations and the corresponding continuous actions as input to estimate the Q value of all possible discrete actions for entity n, and takes the discrete action with the highest Q value as the output, which can be given as

$$K_n^{U_m} = \arg\max_{K \in K_n} Q_n^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}, K; \theta\right), \tag{31}$$

where $Q_n^{U_m}(;\theta)$ denotes the critic network employed for entity n, θ is the set of parameters of the critic network. $\mathcal{O} = \{O^{U_m}\}_{m \in \mathcal{M}}$ represents the observation of all UAVs, $\hat{\mathbf{u}}_{all}^{U_m} = \{\hat{\mathbf{u}}_n^{U_m}\}_{n \in \mathcal{N}_m}$ is the set of the continuous actions regarding all entities.

[2]For convenience of description, we use the term "entity n" to denote ground users and other UAVs that directly communicate with UAV m, i.e., $n \in \mathcal{N}_m = \mathcal{N}_m = \{g, U_k\}_{g \in \mathcal{G}_m, k \in \mathcal{M} \setminus \{m\}} = \{1, 2, ..., N_m, \ldots, N_m + M - 1\}$.

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The final continuous actions of semantic resource allocation can be obtained by

$$\mathbf{u}_n^{U_m} = \hat{\mathbf{u}}_n^{U_m}\left[K_n^{U_m}\right]. \tag{32}$$

In the update of the actor-network, the actor aims to maximize cumulative rewards through action decisions for each observation. The actor-network parameters are updated by maximizing the Q function [50]. For the trajectory optimization subagent, the Q function used to update the actor can be expressed as $Q_{tra}^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}; \theta\right)$. The policy gradient for updating the actor is given by

$$\nabla J_{tra}^{U_m}(\beta) = \left[\nabla Q_{tra}^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}; \theta\right)\nabla \mu_{tra}^{U_m}\right]\Big|_{a,\hat{\mathbf{u}}=\mu(O^{U_m};\beta)}. \tag{33}$$

For the subagent applied to entity n, as shown in part III of Fig. 3, since the final continuous actions of semantic resource allocation correspond only to the optimal discrete action, the Q-function is defined as the Q value of the optimal discrete action, which is represented as

$$MQ_n^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}; \theta\right) = \max_{K\in K_n} Q_n^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}, K; \theta\right). \tag{34}$$

Then, the policy gradient of the actor-network parameters is expressed as

$$\nabla J_n^{U_m}(\beta) = \mathbb{E}\left[\nabla MQ_n^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}; \theta\right)\nabla \mu_n^{U_m}\right]\Big|_{a,\hat{\mathbf{u}}=\mu(O^{U_m};\beta)}. \tag{35}$$

During the update phase of the critic network, the goal of the critic network is to approximate the optimal Q function as the same as possible. For the trajectory optimization subagent, Mean Squared Bellman Error (MSBE) is used as the loss function to measure the proximity between the critic network and the optimal Q-function, which is defined as

$$L_{tra}^{U_m}(\theta) = \left[\left(y_{tra}^{U_m} - Q_{tra}^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}; \theta\right)\right)^2\right], \tag{36}$$

where $y_{tra}^{U_m}$ represents the target value for the current iteration and is given by

$$y_{tra}^{U_m} = r + Q_{tra}^{U_m}(\mathcal{O}', a_{tra}'^{U_m}, \hat{\mathbf{u}}_{all}'^{U_m}; \theta^-), \tag{37}$$

where $\mathcal{O}' = \{O'^{U_m}\}_{m\in\mathcal{M}}$ is the next observation of all UAVs, $a_{tra}'^{U_m}$ and $\hat{\mathbf{u}}_{all}'^{U_m}$ represent the action from the target actors given the next observation.

For the subagent applied to entity n, The critic is the combination of DQN [51], [52] and the critic part of DDPG, which serves two purposes, namely, evaluating the actor's actions for updates and functioning as a DQN to output the discrete action with the maximum Q value. Similar to the trajectory optimization subagent, the loss function of the critic network is expressed as

$$L_n^{U_m}(\theta) = \mathbb{E}\left[\left(y_n^{U_m} - Q_n^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}, K_n^{U_m}; \theta\right)\right)^2\right], \tag{38}$$

where target value of the current iteration $y_n^{U_m}$ is given by

$$y_n^{U_m} = r + Q_n^{U_m}(\mathcal{O}', a_{tra}'^{U_m}, \hat{\mathbf{u}}_{all}'^{U_m}, K_n'^{U_m}; \theta^-), \tag{39}$$

where $K_n'^{U_m} = \arg\max_{K'\in\mathcal{K}_n} Q_n^{U_m}\left(\mathcal{O}', a_{tra}'^{U_m}, \hat{\mathbf{u}}_{all}'^{U_m}, K'; \theta\right)$ is the optimal discrete action given the next observation.

Algorithm 1 Multi-UAV Hybrid Decision Controlled DRL Based Multi-Modal Semantic Communication Resource Allocation (MHDCD-MSC)

Initialize the parameters of the networks.
Initialize replay buffer $\mathcal{R}$.
for `each episode` **do**
  Initialize the environment state of the UAV semantic communication system.
  Receive current states $\mathcal{O} = \{O^{U_m}\}_{m\in\mathcal{M}}$.
  for `each step`t **do**
    for `each UAV agent`U_m **do**
      Obtain action $a_{tra}^{U_m} = \mu_{tra}^{U_m}\left(O^{U_m};\beta\right) + \rho$.
      for `each entity`n **do**
        Obtain action $\hat{\mathbf{u}}_n^{U_m} = \mu_n^{U_m}\left(O^{U_m};\beta\right) + \rho$.
      end for
      for `each entity`n **do**
        With probability ε select random discrete action $K_n^{U_m}$, otherwise: $K_n^{U_m} = \arg\max_{K\in K_n} Q_n^{U_m}\left(\mathcal{O}, a_{tra}^{U_m}, \hat{\mathbf{u}}_{all}^{U_m}, K; \theta\right)$.
        Select continuous action: $P_n^{U_m}, B_n^{U_m}, \mathbf{u}_n^{U_m} = \hat{\mathbf{u}}_n^{U_m}\left[K_n^{U_m}\right]$.
      end for
    end for
    Execute action and calculate reward r and obtain next environment state $\mathcal{O}' = \{O'^{U_m}\}_{m\in\mathcal{M}}$.
    Store tuple$(O^{U_m}, a_{tra}^{U_m}, \hat{\mathbf{u}}_n^{U_m}, K_n^{U_m}, r, O'^{U_m})_{m\in\mathcal{M}, n\in\mathcal{N}_m}$ into replay buffer $\mathfrak{D}$.
    Randomly sample a mini-batch tuples with size D_b from the replay buffer $\mathfrak{D}$.
    for `each UAV agent`U_m **do**
      Update actor network using $\nabla J_{tra}^{U_m}(\beta)$
      Update critic network using $L_{tra}^{U_m}(\theta)$.
      for `each entity`n **do**
        Update actor network using $\nabla J_n^{U_m}(\beta)$.
        Update critic network using $L_n^{U_m}(\theta)$.
      end for
    end for
    Soft update target networks of critic and actor using $\theta^- \leftarrow \tau\theta + (1-\tau)\theta^-$, $\beta^- \leftarrow \tau\beta + (1-\tau)\beta^-$.
  end for
end for

The proposed resource allocation scheme for the multi-modal semantic communication UAV collaborative network is outlined in Algorithm 1.

V. Simulation Results

In this section, extensive simulations are conducted to evaluate the performance of our proposed semantic-aware intelligent resource allocation scheme for the multi-modal semantic communication UAV collaborative networks.

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A. Simulation Setup

We consider a UAV semantic communication network composed of $M = 2$ regions, each with a radius of $200m$ and separated by a distance of $600m$ from each other. Within each region, the flight altitude of the UAV is fixed at $Z = 100m$, and the initial horizontal coordinate of the UAV is at the center of the region. The maximum flight distance of the UAV is $l_{\max} = 30m$. Each region has $N_m = 5$ randomly distributed ground users, and each user moves $10m$ in a random direction during each time slot. Additionally, 2 ground users receive information from the local region, while the remaining 3 ground users receive information from another region. Each UAV is equipped with the maximum bandwidth of $B_{\max} = 2.5MHz$ and the maximum power of $P_{\max} = 6W$. The size of the collected image information, denoted as $A^{U_m}(t)$, is randomly selected from the range $[200, 400]$ patchs. The growth rates of service satisfaction are $\zeta_D = 5$ and $\zeta_H = 0.25$. The transmission cost weighted parameters are $\alpha_p = \alpha_b = 0.5$. The weighted parameters of QoE and transmission cost are $\alpha_q = 0.8$ and $\alpha_c = 0.2$, respectively. The threshold of the transmission accuracy is $\phi_{th} = 0.5$. Additionally, the channel-related parameters include $\eta_a = 12.08$, $\eta_b = 0.11$, $\varphi_{NLOS} = 1.6\ dB$, $\varphi_{NLOS} = 23dB$, $v_1 = 1$,$v_2 = 0.05$, and the noise power spectral density is $n_0 = 10^{-17}$ W/Hz. The aforementioned parameters are commonly used in works [36], [44].

In the proposed multi-modal semantic communication system, the semantic encoders for both the image-to-image transceiver and the image-to-text transceiver are implemented by pre-trained ViT models [38], [39]. The dimensions of the input image are 224×224, while the patch size is 16×16. Their semantic decoders are composed of 8 and 4 layers of typical Transformer decoding layers, respectively. The "Karpathy splits" dataset [39], which includes 123287 images with 5 related text descriptions for each image, is employed for system training and testing. The sets of optional numbers of the transmitted semantic symbols for image and text transmission are $\mathcal{K}_I = \{192, 256, 320, 384\}$ and $\mathcal{K}_T = \{36, 48, 60, 72\}$, respectively. The initial parameters of the fitting function model $[L_{K_J}, \gamma_{K_J}, \zeta_{K_J}, b_{K_J}] = [1, 0, 1, 0]$. Furthermore, the accuracy constraint for the approximate semantic entropy is $\nu = 0.8$.

For the implementation of the MHDCD-MSC scheme, the actor and critic networks of each subagent both consist of two hidden layers. The first hidden layer has 400 neurons, while the second hidden layer has 300 neurons. The learning rates for the actor and critic networks are set as 0.0001 and 0.001, respectively. The soft update rate for target networks is set as $\tau = 0.001$, while the discount factor is set as $\gamma = 0.99$. Additionally, an ε-greedy strategy with a value of ε=0.1 is utilized. The sizes of the mini-batch and replay buffer for training are set as 256 and 20000, respectively. The scheme is trained over 500 episodes, with each episode containing 200 time steps.

To validate the effectiveness of MHDCD-MSC, we compare it with five benchmark schemes:

1) *MHDCD-MSC With Fixed Position (MHDCD-MSCFP):* The UAV adopts the multi-modal semantic transmission method with fixed position. This system leverages the proposed MHDCD scheme to optimize communication resource allocation in terms of the transmit power, the communication bandwidth, and the number of the transmitted semantic symbols.
2) *MHDCD Based Image Semantic Communication Resource Allocation (MHDCD-ISC):* The semantic communication only adopts the image-to-image transceiver to transmit information with image modality. This system leverages the proposed MHDCD scheme to optimize communication resource allocation in terms of the UAV trajectory, the transmit power, the communication bandwidth, and the number of the transmitted semantic symbols. This reflects the scheme in [27], which introduces semantic optimization factors for single-modal image transmission.
3) *MADDPG Based Image Semantic Communication Resource Allocation With Fixed Number of Semantic Symbols (MADDPG-ISCFS)):* The semantic communication adopts the image-to-image transceiver to transmit information with image modality, the number of the transmitted semantic symbols is fixed as 256. This system leverages the the multi-agent deep deterministic policy gradient (MADDPG) [49] to optimize communication resource allocation in terms of the UAV trajectory, the transmit power, and the communication bandwidth. This scheme is aligning with the approach in [31], which focuses solely on conventional optimization factors without considering semantic optimization factors.
4) *MADDPG Based Traditional Communication Resource Allocation (MADDPG-TC):* The traditional communication only transmits image modality, where the source encoder utilizes Joint Photographic Experts Group (JPEG) and Better Portable Graphics (BPG) encoding [14]. The channel encoder employs Low-Density Parity-Check (LDPC) codes. This system adopts the MADDPG scheme to optimize the UAV trajectory and the allocation of the transmit power and bandwidth.
5) *Equal Allocation Based Multi-Modal Semantic Communication Resource Allocation (EA-MSC):* The UAVs adopt the multi-modal semantic transmission method with fixed position. Both bandwidth and transmit power are equally allocated to each user and UAV, and the number of the transmitted semantic symbols is optimized by the DQN scheme.

B. Performance Evaluation

Fig. 4 and 5 show the transmission accuracy of the proposed multi-modal semantic communication model versus the SNR under different semantic transmission symbol numbers. Here, each point corresponds to actual measured values at different SNRs, while the curves are obtained by the sigmoid curve-fitting based on the actual measured values, representing the accuracy functions $f_J(K_J, \gamma)$, $J \subseteq \{I : image,\ T : text\}$. From Fig. 4 and 5, it is evident that the transmission accuracy of both the image-to-image transceiver and image-to-image semantic transceiver tends to decrease at lower SNR values. It is worth noting that the decrease of the transmission accuracy

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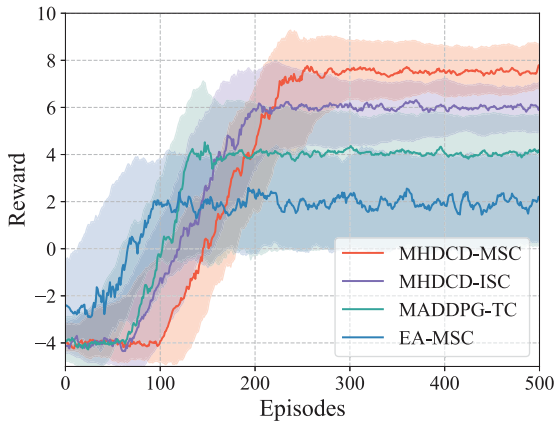


Fig. 6. The convergence of different resource allocation schemes.

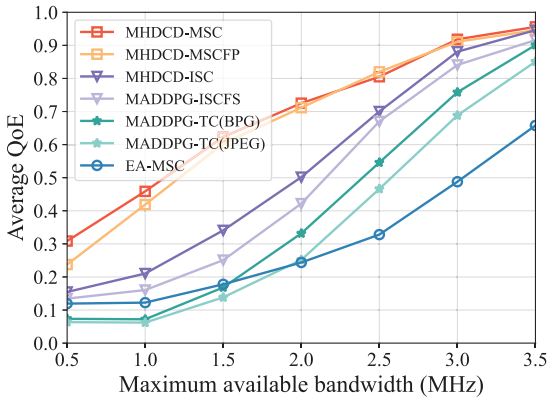


Fig. 7. Average QoE with different maximum available transmit bandwidths.

for each semantic transceiver becomes more pronounced when the number of the semantic transmission symbols is fewer. This decrease is largely due to the fact that a lower number of semantic transmission symbols implies a higher degree of compression of the original information and a reduced error-correcting capability. The data that is highly compressed is typically more sensitive to transmission errors, leading to the decrease of the transmission accuracy in environments with low SNR. Conversely, a higher number of semantic transmission symbols not only preserves more of the original data but also enhances the error-correcting capabilities. This results in robust performance across diverse channel conditions. Therefore, in the optimization of resources for the semantic communication system, it is imperative to consider not only transmit power and bandwidth, closely related to SNR, but also the transmission of semantic symbols at the semantic level.

Fig. 6 shows the convergence of the proposed semantic-aware resource allocation scheme, MHDCD-MSC, compared to the three main baseline schemes. The shaded areas represent the range of reward fluctuations, while the solid lines denote the smoothed reward values. As the number of iterations increases, the reward values of all methods exhibit a steady convergence trend. In terms of the convergence speed, the proposed MHDCD-MSC is slightly slower than the other three baselines. This is primarily because MHDCD-MSC has a larger action space and higher computational complexity. However, in terms of the converged reward values, MHDCD-MSC achieves noticeably the largest rewards among the other baseline schemes. The rewards obtained by the semantic communication-based MHDCD-MSC and MHDCD-ISC surpass those achieved by the traditional communication-based MADDPG-TC. This indicates that semantic communication can achieve a better QoE-cost tradeoff than the conventional communication in UAV communication networks with limited resources. The reward value of MHDCD-MSC using multi-modal semantic communication is higher than that of MHDCD-ISC with single-modal transmission, which indicates that multi-modal semantic communication models can further improve the performance of UAV communication networks. Additionally, Compared with EA-MSC, which also employs multi-modal semantic transmission but utilizes equal resource allocation, MHDCD-MSC achieves a significantly higher reward, verifying the effectiveness of the resource allocation scheme adopted in MHDCD-MSC. Besides, although EA-MSC only optimizes the number of semantic symbols transmitted, its modest reward improvement highlights the critical role of the optimization of semantic symbols in the semantic communication resource optimization.

Fig. 7 shows the average QoE versus the maximum available transmit bandwidth under different schemes. It can be seen that the average QoE for each scheme increases with the maximum bandwidth. This phenomenon is attributed to the fact that a larger transmission bandwidth reduces transmission delays and accommodates more transmission symbols, thereby enhancing the average QoE of ground users. Within the low-bandwidth range (i.e.,0.5–1.5 MHz), the proposed MHDCD-MSC achieves a higher QoE than MHDCD-MSCFP, where the positions of UAVs are fixed. This improvement stems from trajectory optimization, which effectively enhances the channel environment. Semantic communication, which focuses on transmitting meaningful information while eliminating redundancy, further contributes to performance gains. As a result, the average QoE obtained by MHDCD-MSC and MHDCD-ISC based on semantic communication is consistently higher than that of MADDPG-TC based on the traditional communication. Among the traditional communication schemes, MADDPG-TC(BPG) achieves better QoE than MADDPG-TC(JPEG), attributed to the superior compression efficiency of BPG encoding. Similarly, multimodal semantic communication in MHDCD-MSC provides a significant advantage over the single-modal MHDCD-ISC. This advantage arises from the ability of multimodal communication to enable UAVs to adjust transmission modalities(i.e., text), thereby reducing data volume and transmission time. For similar reasons, EA-MSC outperforms MADDPG-TC in QoE under low-bandwidth conditions, as it benefits from the adjustment of semantic modality. However, when the available bandwidth becomes relatively sufficient, MADDPG-TC gradually surpasses EA-MSC. This is because MADDPG-TC leverages DRL-based resource allocation strategies, enabling more efficient use of the available bandwidth. Finally, compared to MADDPG-ISCFS with the fixed number of semantic

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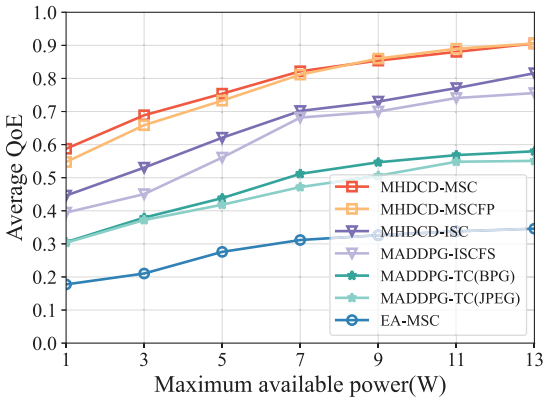


Fig. 8. Average QoE with different maximum available transmit powers.

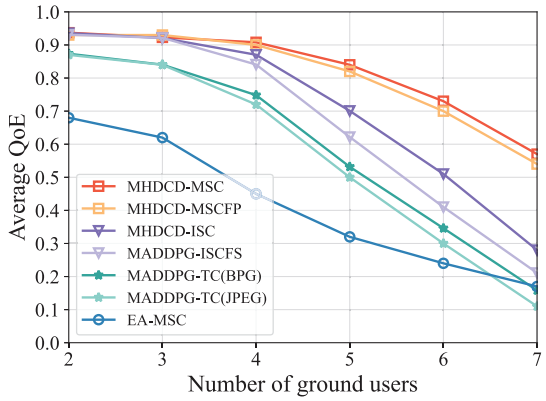


Fig. 9. Average QoE with the different numbers of ground users.

symbols, MHDCD-ISC achieves higher QoE by optimizing semantic symbols. This finding highlights the importance of semantic-level optimization in improving the overall performance of semantic communication systems.

Fig. 8 shows the average QoE versus the maximum available transmit power under different schemes. Notably, as a higher transmit power effectively enhances the transmission rate, leading to the increased volume of the transmitted semantics, the average QoE of all schemes experiences growth with the maximum transmit power. Among the schemes, the proposed MHDCD-MSC achieves the highest average QoE, while MHDCD-MSCFP, which does not optimize UAV trajectories, shows a slight reduction in term of QoE. In contrast, EA-MSE exhibits the lowest average QoE. These results further demonstrate that effective trajectory optimization and resource allocation strategy are crucial for ensuring UAV operational quality and alleviating communication stress in scenarios with limited wireless resources. When compared with MADDPG-TC based on the traditional communication, MHDCD-MSC and MHDCD-ISC can achieve higher QoE, which shows that semantic communication can further improve the performance of UAV communication networks. Additionally, MHDCD-ISC achieves higher QoE compared to MADDPG-ISCFS, which further emphasizes the significance of semantic symbol optimization in semantic resource allocation.

Fig. 9 shows the average QoE versus the number of ground users. Intuitively, the average QoE of all methods

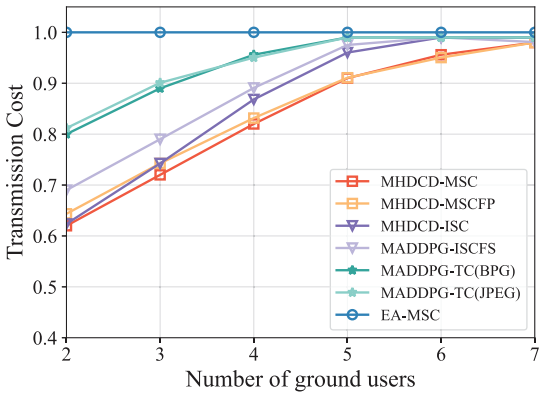


Fig. 10. Transmission cost with the different numbers of ground users.

decreases with the number of ground users while the proposed MHDCD-MSC always achieves the best QoE performance. In comparison, MHDCD-MSCFP, which does not optimize UAV trajectories, experiences a slight reduction in QoE. It is worth noting that when there exist fewer ground users, the QoE of MADDPG-TC surpasses that of EA-MSC due to the more rational resource allocation of the transmit power and bandwidth. However, as the number of ground users becomes larger, the QoE benefits more from semantic communication than from efficient resource allocation. Thus, the QoE of the semantic-based EA-MSC is better than that of the traditional bits-based MADDPG-TC, which further demonstrates that semantic communication can achieve superior performance compared to the traditional communication in low bandwidth and the low power situations. Additionally, MADDPG-ISCFS, which does not optimize the number of semantic symbols, exhibits slightly lower QoE than MHDCD-ISC. This result highlights the critical role of semantic symbol optimization in enhancing the overall performance of semantic communication systems.

Fig. 10 shows the UAV transmission cost versus the number of ground users. Since EA-MSC adopts an equal allocation method for communication resources, its transmission cost remains constant being 1. When the number of ground users is relatively low and available communication resources are relatively abundant, MHDCD-MSC, MHDCD-ISC, and MADDPG-TC effectively reduce transmission costs through a rational allocation of communication resources facilitated by the DRL-based resource allocation strategies. Moreover, due to the focus on the original meaning of information and the reduction of data transmission through semantic extraction, the transmission costs of MHDCD-MSC and MHDCD-ISC are lower than those of MADDPG-TC. Benefiting from trajectory optimization, MHDCD-MSC exhibits slightly lower transmission costs than MHDCD-MSCFP. Among traditional communication schemes, MADDPG-TC(BPG) achieves slightly lower transmission costs than MADDPG-TC(JPEG) due to its higher source compression rate. Besides, the optimization of semantic symbol transmission allows MHDCD-ISC to achieve slightly lower costs than MADDPG-ISCFS. With a large number of users and limited communication resources, the transmission cost of MHDCD-MSC remains lower than those

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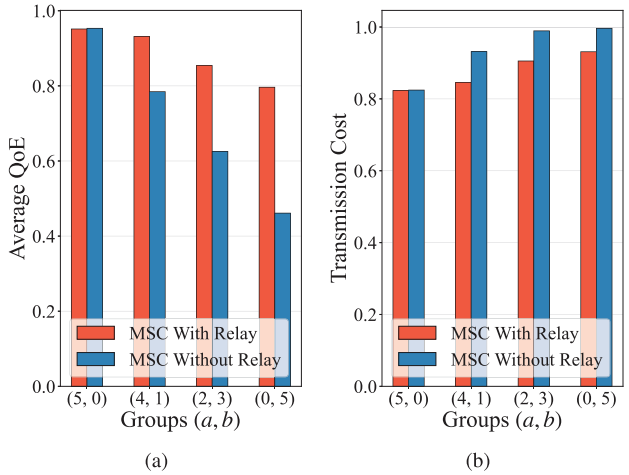


Fig. 11. (a) Average QoE and (b) Transmission cost versus the different groups.

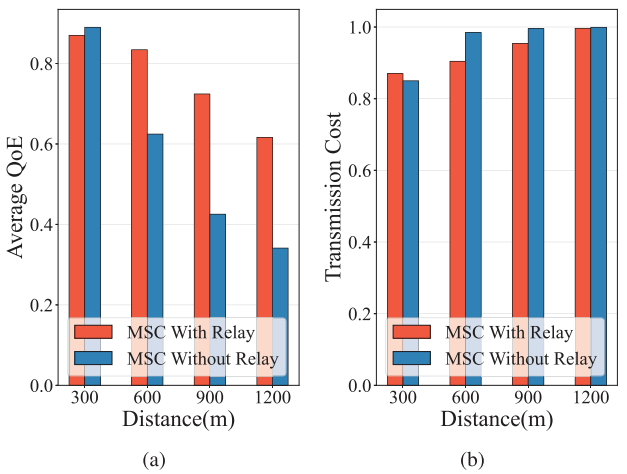


Fig. 12. (a) Average QoE and (b) Transmission cost versus the different distances.

of other benchmark schemes, which further emphasizes that multi-modal semantic communication can achieve superior performance of the semantic communications where communication resources are scarce.

Fig. 11(a) and Fig. 11(b) compare the average QoE and transmission costs of UAV semantic communication networks with and without relay communication in different user group scenarios. The user group is denoted by the tuple (a, b), where a denotes the number of ground users receiving information from the local region, and b denotes the number of ground users receiving information from adjacent regions. When ground users in each region solely receive local information, i.e.,$(5, 0)$, both strategies achieve identical average QoE and transmission costs due to the absence of inter-region communication, which requires only direct communication. As the number of ground users requesting the images from the adjacent region gradually increases, the average QoE for both strategies declines, while their transmission costs rise. However, UAV communication networks with relay communication outperform those without relay communication in terms of both QoE and transmission costs. This performance gap is primarily attributed to the increased communication challenges posed by the growing number of long-distance communication links in non-relay UAV semantic communication networks. These challenges lead to higher transmission costs and degraded communication quality. By taking advantage of UAV cooperative relaying transmission, such challenges can be effectively addressed, and the communication quality can be enhanced.

Fig. 12(a) and Fig. 12(b) illustrate the average QoE and transmission costs of UAV semantic communication networks with and without relay communication under different distances between regions. When the average distance between regions is relatively short, i.e., $300m$, the A2G channel conditions between UAVs and ground users are relatively favorable. In this case, additional relay communication introduces unnecessary resource and time overhead, making non-relay communication networks superior in terms of both QoE and transmission cost. However, as the average

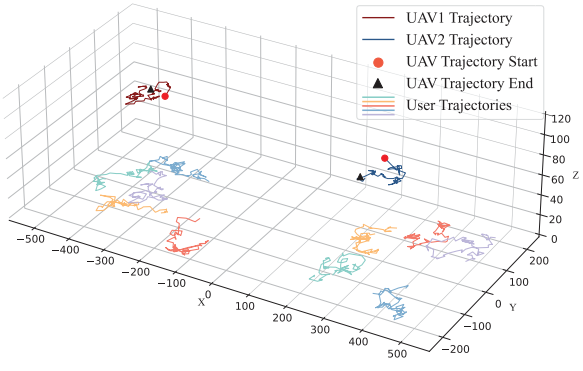


Fig. 13. The UAV trajectories of MSC with relay.

distance between regions increases, the average QoE of non-relay communication networks declines significantly, while transmission costs rise steadily. This deterioration is primarily due to increased path loss and communication obstacles between the air and ground. Furthermore, when the distance becomes excessively long, i.e., $1200m$, both non-relay and relay communication networks utilize all available resources to compensate for the severe path loss resulting in a transmission cost of 1 for both. Nevertheless, the proposed UAV cooperative relay communication leverages A2A transmission to maintain superior communication performance and reduce transmission costs. These results demonstrate that cooperative relay communication can effectively enhance the performance of UAV semantic communication networks.

Fig. 13 and 14 depict the UAV trajectories in UAV semantic communication networks with and without relay communication. In UAV semantic communication networks with relay communication, UAVs benefit from cooperative relay communication, allowing each UAV to focus on providing high-quality communication for local users. As a result, they tend to hover near the center of their respective regions. In contrast, in UAV semantic communication networks without relay communication, UAVs must ensure adequate communication quality for users in the adjacent region. Consequently,

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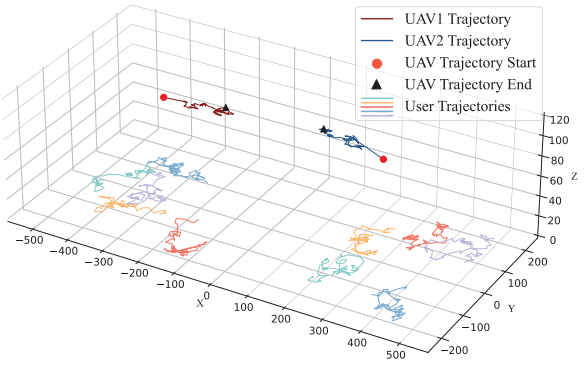


Fig. 14. The UAV trajectories of MSC without relay.

each UAV moves closer to the adjacent region, which not only increases unnecessary flight costs but also degrades the QoE of overall communication network, which is illustrated in Fig. 11(a) and 12(a). These findings further highlight the critical role of cooperative relay communication in enhancing the communication efficiency and performance of UAV semantic communication networks.

VI. Conclusion

In this paper, we studied a semantic-aware resource allocation for multi-modal semantic communication in UAV collaborative networks. To enhance the efficacy of UAV semantic communication, a novel multi-modal semantic communication system was proposed to transmit information with different semantic modalities. A semantic-level resource allocation problem was formulated to maximize QoE while reducing transmission costs by optimizing the UAV trajectory, the spectrum bandwidth, the transmit power and the number of the transmitted semantic symbols. To tackle the dynamic non-convex problems, a hybrid MHDCD-MSC scheme was proposed to handle the challenging hybrid discrete and continuous actions. Our simulation results demonstrate the effectiveness of our proposed resource allocation scheme and show the superiority of the UAV collaborative relay communication combined with multi-modal semantic communication over baseline methods.

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